Building Materials

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Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood
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Chapter Four：Concrete

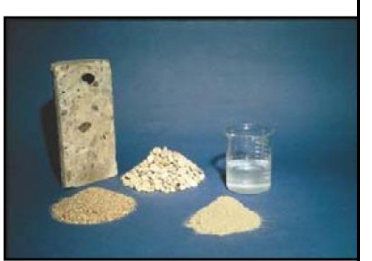
1. Introduction
2. Compositions
3. Structure
4. Strength and Damage
5. Mix Proportioning
6. Early-age Properties
7. Volume Changes
8. Durability

4.1 Introduction

What is concrete?

Concrete is an artificial stone made essentially of Portland cement, water, and aggregates.

- Mix of water, cement, sand, aggregates, mineral additions, air and admixtures
- paste= water + fines (cement, fillers, slag, silica fume, fly-ash, etc)
- mortar = paste + sand (<5mm)
- concrete = mortar + aggregate (> 5mm)



4.1 Introduction

What is concrete?

- Most consumed material only after water
  - Annual usage 7 billion m^3
  - More than 1 m^3 / person / year



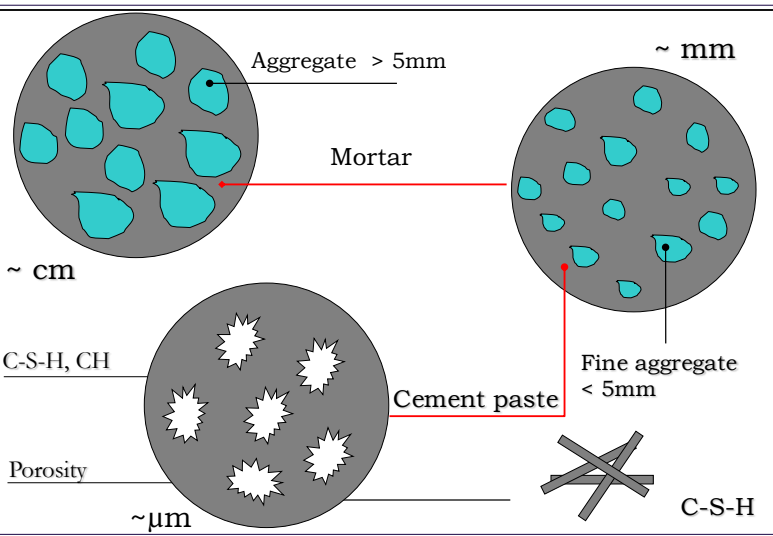


- Used for over 2000 years in structures from Roman concrete to modern bridges, highways, and buildings



- Vast amounts of natural resources (limestone, clay, aggregates, sands, water) needed for making cement and concrete
- Cement production responsible for 5 - 7% world CO_2 emission

4.1 Introduction



4.1 Introduction

General about concrete

- Strong in compression,
- Inexpensive,
- Easy to make.







Advantages:

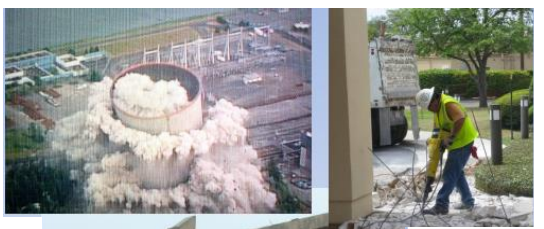
- Ability to be cast
- Economical
- Durable
- Fire resistant
- On-site fabrication

Disadvantages

- Low tensile strength
- Low ductility
- Volume instability
- Low strength to weight ratio

4.1 Introduction

Concrete is Recyclable









Reinforcement is collected



What are the recycled concrete used for?

4.1 Introduction

Green Concrete

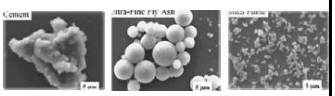
- How to Make Concrete Greener?

- Use more concrete, but less cement

- Less cement with SCMs (Supplementary Cementitious Materials)
  - Replacing up to 30 – 50% Portland cement
    - Fly ash, slag, Silica fume, MK
    - Recycled glass powder

- More recycled materials in concrete
  - As aggregate and fillers used in concrete up to 80%
  - Recycled concrete
  - Recycled glass, tire scraps, plastics
  - Limestone powder





4.1 Introduction

Types of Concrete

Strength:
- low strength concrete (compressive strength< 20 MPa)
- Medium strength concrete (compressive strength 20-40 MPa)
- High strength concrete (compressive strength > 40 MPa)

Where it is used:
- Plain Concrete
- Reinforced Concrete
- Prestressed Concrete

Density:
- lightweight concrete (density < 1 950 kg/m^3)
- normal-weight concrete(density ~ 2 400 kg/m^3)
- over-weight concrete(density > 2 600 kg/m^3)

4.1 Introduction

Types of Concrete

Ways of construction:

- pumping concrete
- shotcrete
- Slipform paving concrete
- Roller compacting concrete
- Self-compacting concrete
- …

Production:

- Precast
- Cast-in-Place
  - ✓ Prefabricating in plant
  - ✓ Mixing in situ







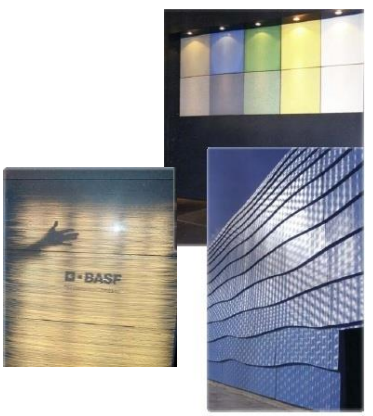
4.1 Introduction

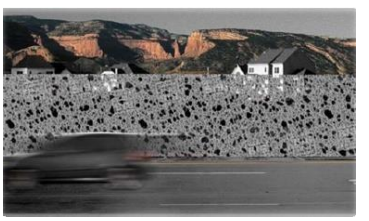


New Technology

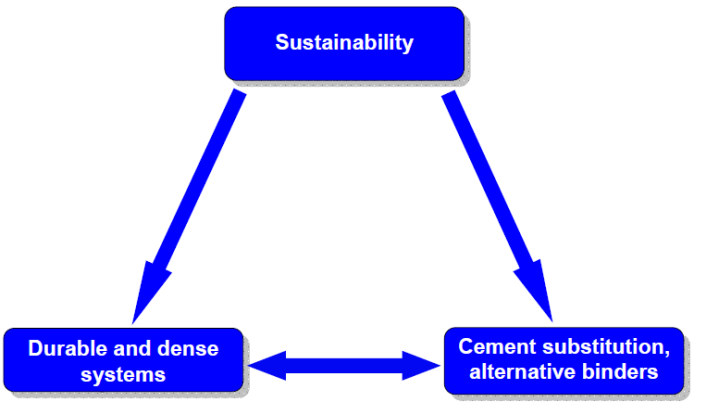
- World of Concrete Beyond Structural





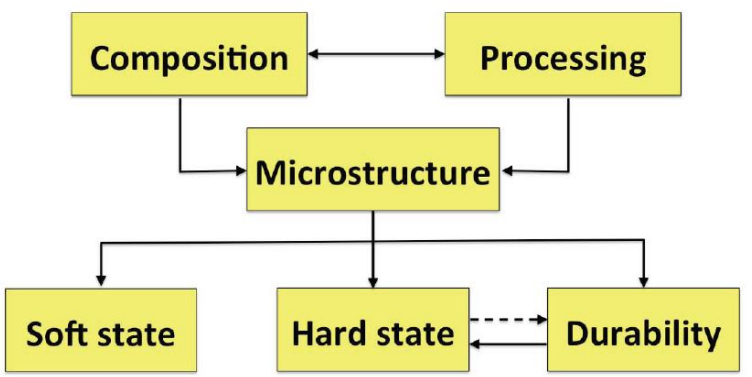


Challenges for the future



4.1 Introduction

Materials science & civil engineering

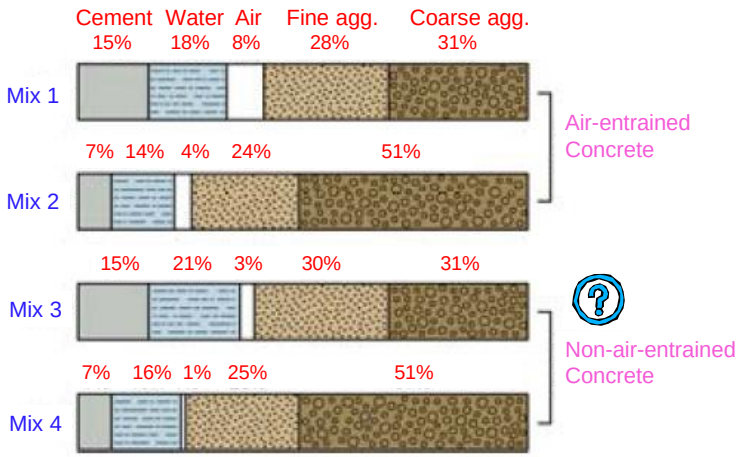


Chapter Four：Concrete

1. Introduction
2. **Compositions**
3. Structure
4. Strength and Damage
5. Mix Proportioning
6. Early-age Properties
7. Volume Changes
8. Durability

4.2 Compositions

Range in proportions of materials used in concrete, by absolute volume.



4.2 Compositions

Cement

basic elements



4.2 Compositions

Cement

- It was named Portland cement because it looked like the stone quarried on the Isle of Portland.



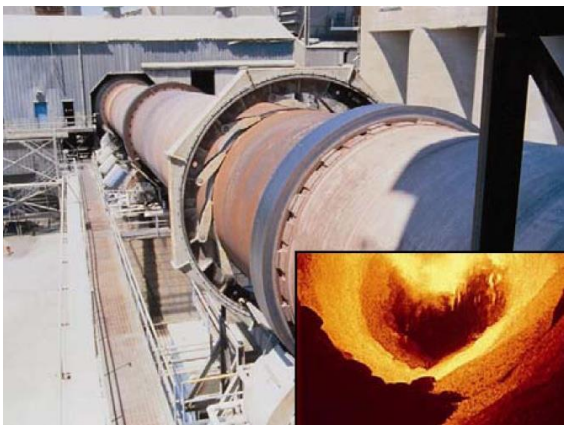


- Joseph Aspdin (1779-1835) patented the clay and limestone cement known as Portland cement in 1824. Joseph's son, William Aspdin used to make the first genuine Portland cement.
- Portland cement was first used in the civil engineering project by Isambard Kingdom Brunel (1806-1859), as the lining of the Thames Tunnel.

4.2 Compositions

Cement

- Portland cement is produced by mixing ground limestone, clay or shale, sand and iron ore.
- This mixture is heated in a rotary kiln to temperatures as high as 1,600 degrees Celsius.
- The heating process causes the materials to break down and recombine into new compounds (clinker) that can react with water in a crystallization process called hydration.



4.2 Compositions

Cement

Clinker

components	content
Tricalcium silicate (C_3S)	55~65%
Dicalcium silicate (C_2S)	15~25%
Tricalcium aluminate (C_3A)	8~14%
Tetracalcium aluminoferrite (C_4AF)	8~12%

$C = CaO$ $S = SiO_2$

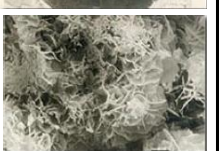
$A = Al_2O_3$ $F = Fe_2O_3$

$H = H_2O$ $\hat{S} = SO_3$

C_3S 

C_2S 

C_3A 

C_4AF 

4.2 Compositions

Cement

$C = CaO$ $S = SiO_2$

$A = Al_2O_3$ $F = Fe_2O_3$

$H = H_2O$ $\hat{S} = SO_3$

Mineralogical Composition of Classic Cement Clinker

Oxide Composition	Cement Notatiion	Common Name	Concentration (wt%)
$3CaO \cdot SiO_2$	C_3S	alite	55–65
$2CaO \cdot SiO_2$	C_2S	belite	15–25
$3CaO \cdot Al_2O_3$	C_3A	aluminate	8–14
$4CaO \cdot Al_2O_3 \cdot Fe_2O_3$	C_4AF	brownmillerite	8–12

4.2 Compositions

Cement

Density:

3000 kg/m^3~3150 kg/m^3

Fineness:

300 m^2/kg~350 m^2/kg

4.2 Compositions

Cement Hydration

- When mixing the water and cement constitute a paste which surrounds all the individual pieces of aggregate to make a plastic mixture.
- A chemical reaction called *hydration* takes place between the water and cement, and concrete normally changes from a plastic to a solid state in about 2 hours.
- Heat of hydration - is the heat given off during *the* chemical reaction as the cement hydrates.

components	content
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Dicalcium silicate (C_2S)	15~25%
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4.2 Compositions

Cement Hydration

- Hydration of tricalcium silicate (C_3S) and dicalcium silicate (C_2S) produce calcium silicate hydrates (C-S-H), calcium hydroxide (CH) and heat:

Reaction of C_3S: $2C_3S + 6H \rightarrow C_3S_2H_3 + 3CH$ (1)

CSH

Reaction of C_2S: $2C_2S + 4H \rightarrow C_3S_2H_3 + CH$ (2)

- ✓ Both produce C-S-H and CH as reaction products
- ✓ C_2S produces less CH (important for durability in sulfate rich environments)
- ✓ More heat is evolved during C_3S hydration
- ✓ C_3S hydration is more rapid, contributing to early age strength (2-3h to 14 days)
- ✓ C_2S hydration occurs more slowly to contributing to strength after ~14 days.

4.2 Compositions

Cement

Cement types	ordinary	early-strength	low heat	sulfate resistant
C_3S	48	65 ↑	31 ↓	42 ↓
C_2S	24	11	40	34
C_3A	13	8	12	2 ↓
C_4AF	9	9	12	15



4.2 Compositions

Cement

ACI Committee 116 defines mass concrete as :

"any large volume of cast-in-place concrete with dimensions large enough to require that measures be taken to cope with the generation of heat and attendant volume change to minimize cracking."





4.2 Compositions

Cement Hydration

- Reaction of C_3A:

In a system without lime and gypsum

$$2C_3A + 21H \rightarrow 2C_3AH_6 + 9H \qquad (3)$$

4.2 Compositions

Cement Hydration

- Reaction of C_3A:

When lime is present (in the absence of gypsum)

$$C_3A + CH + 12H \rightarrow C_4AH_{13} \qquad (4)$$

Flash set

When both lime and gypsum ($C\bar{S}H_2$) are present, as in the real case of cement hydration, in addition to eq. (4) the following reactions take place:

ettringite

$$C_4AH_{13} + 3C\bar{S}H_2 + 14H \rightarrow C_6A\bar{S}_3H_{32} + CH \qquad (5)$$

$$C_6A\bar{S}_3H_{32} + 2C_4AH_{13} \rightarrow 3C_4A\bar{S}H_{12} + 2CH + 20H \qquad (6)$$

More stable phase of monosulphate

Ettringite deposited on the surface of the hydrating C_3A, provides an effective barrier against further rapid hydration

By combining Eq. (4 - 6):

$$C_3A + C\bar{S}H_2 + 10H \rightarrow C_4A\bar{S}H_{12} \qquad (7)$$

4.2 Compositions

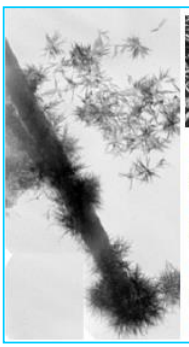
Cement Hydration

- Reaction of C_3A: (Ettringite, Aft)

$$C_4AH_{13} + 3C\bar{S}H_2 + 14H \rightarrow C_6A\bar{S}_3H_{32} + CH \quad (5)$$

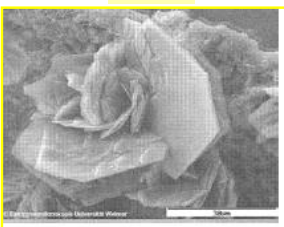
(monosulfate, Afm)

$$C_6A\bar{S}_3H_{32} + 2C_4AH_{13} \rightarrow 3C_4A\bar{S}.H_{12} + 2CH + 20H \quad (6)$$





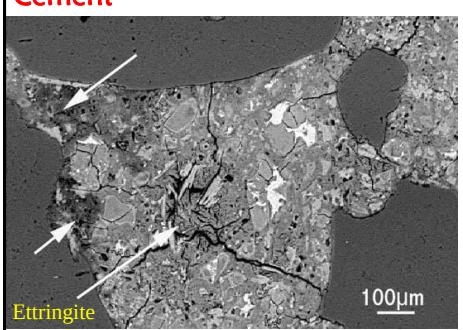
- Needle-like morphology
- Needles interlock, take up much water
- contributes to stiffening of mixture
- some early strength



- hexagonal plate morphology arranged in "rosettes"
- vulnerable to sulfate attack

4.2 Compositions

Cement







4.2 Compositions

Cement

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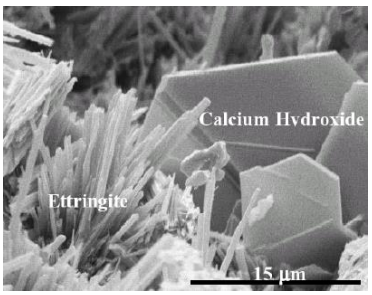
4.2 Compositions

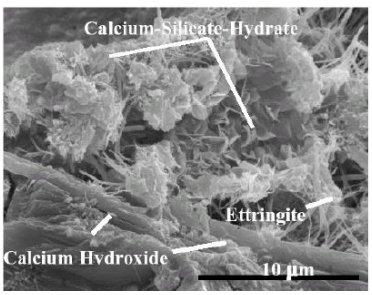
Cement Hydration

Reaction of C_4AF:

C_4AF reacts at a slower rate than C_3A

$$C_4AF + 2CH + 10H \rightarrow C_6AFH_{12} \quad (8)$$





4.2 Compositions

Cement Hydration

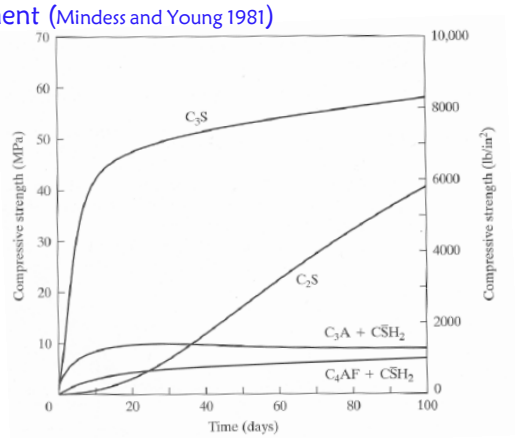
Influence of cement compounds on concrete properties

Property	Compound	remarks
Setting behavior	C_3S	Controls normal setting
	C_3A	Can cause premature stiffening
Temperature rise during hydration	C_3S	
	C_3A	
Strength development	C_3S	Responsible for early strength
	C_2S	Contributes to long-term strength
Creep and shrinkage	C_3S, C_2S	Major contributors
	C_3A, C_4AF	Minor effects
durability	C_3S	Leaching of $Ca(OH)_2$
	C_3A	sulfate attack

4.2 Compositions

Cement Hydration

Strength development over time of the four main chemical components in portland cement (Mindess and Young 1981)



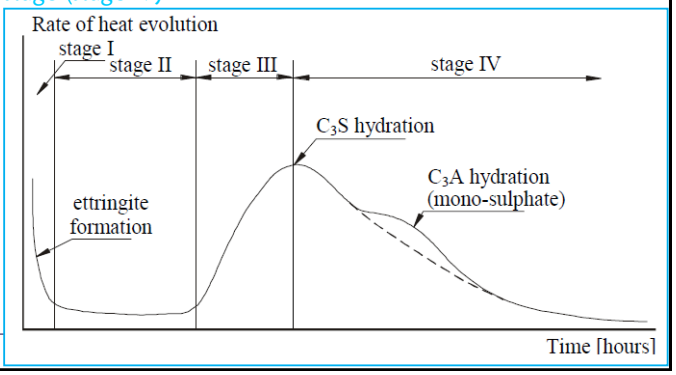
4.2 Compositions

Cement Hydration

Hydration process:

1. Pre-induction period (stage I)
2. Dormant stage (stage II)
3. Acceleration stage (stage III)
4. Post-acceleration stage (stage IV)

Rate of heat evolution during the hydration of Portland cement, (after Mindess & Young [1981])



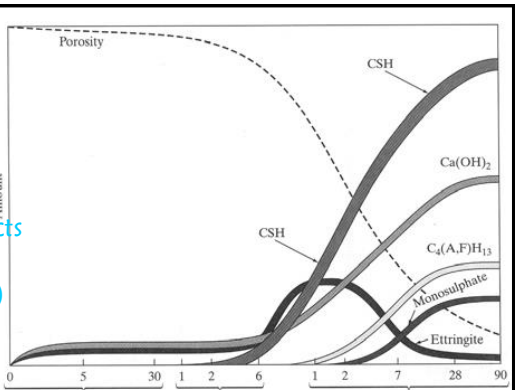
4.2 Compositions

Cement Hydration

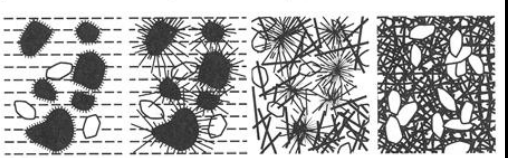
Hydration process:

Formation of hydration products over time leads to:

- Stiffening (loss of workability)
- Setting (solidification)
- Hardening (strength gain)



Evolution of hydration products and pore structure (after Locher [1976])

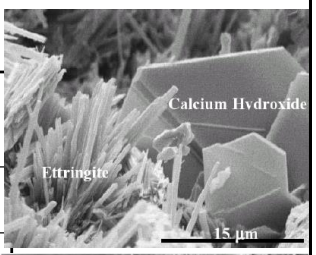


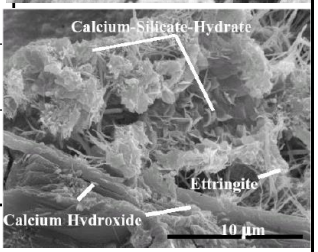
4.2 Compositions

Cement Hydration

Hydration products:

Components	content (%)	density (g/cm^3)	Surface area (m^2/g)	shape
C-S-H	65	2.0	400	Amorphous
CH	20	2.25	0.5	Crystal
$C_4A\underline{S}H_{12}$	10	1.95	2	Thin plate
Unhydrated cement particle	5	3.15	0.1	particle



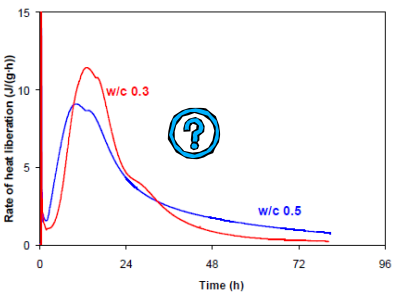


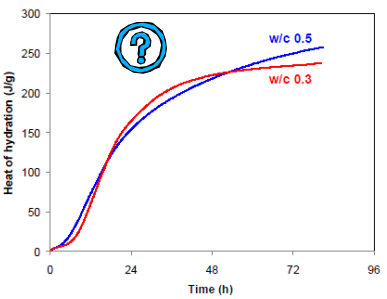
Which one is more porous?

4.2 Compositions

Cement Hydration

Rate of heat liberation measured in isothermal calorimeter





- Typical 400-450 J/g cement reacted, increases with C_3S and C_3A
- Lower heat of hydration with low w/c
- Degree of hydration can be calculated

4.2 Compositions

Cement Hydration

Degree of hydration:

the degree of hydration, α (t) [-], is defined as the ratio of the amount of cement that has reacted at time t relative to the original amount of cement [van Breugel 1991]:

$$\alpha(t) = \frac{amount\ of\ cement\ that\ has\ reacted\ at\ time\ t}{total\ amount\ of\ cement\ at\ time\ t = 0}$$

The degree of heat liberation for practical and research purposes can be used as an indicator of the degree of hydration [van Breugel 1991]:

$$\alpha(t) \approx \alpha(Q(t)) = \frac{Q(t)}{Q_{pot}}$$

4.2 Compositions

Cement Hydration

Types of cement:

- Non-hydraulic cements - hardening will not take place under water. For example: mud (clay).
- Hydraulic cements - Hardening will occur under water. For example: Portland cement, slag.





Summary

1. Concrete is Everywhere

Types of concrete?

Compositions of concrete, advantage and disadvantage?

2. Cement Composition

Production of cement?

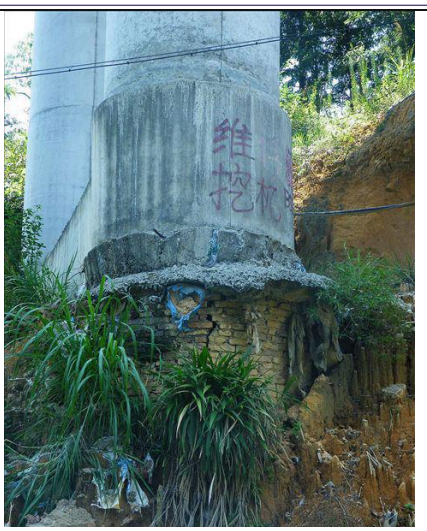
Components of clinker, content, influence on concrete properties?

Why gypsum?

Reaction?

Early-strength, low heat, and sulfate resistant cement?

Degree of hydration, properties of hydration products?



QUESTIONS?